

CS611 Machine Learning Engineering

Lecture 1: Introduction

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Outline

Machine Learning Lifecycle

Challenges of Operationalizing ML

Requirements for Production ML Systems

ML Pipeline and ML Steps

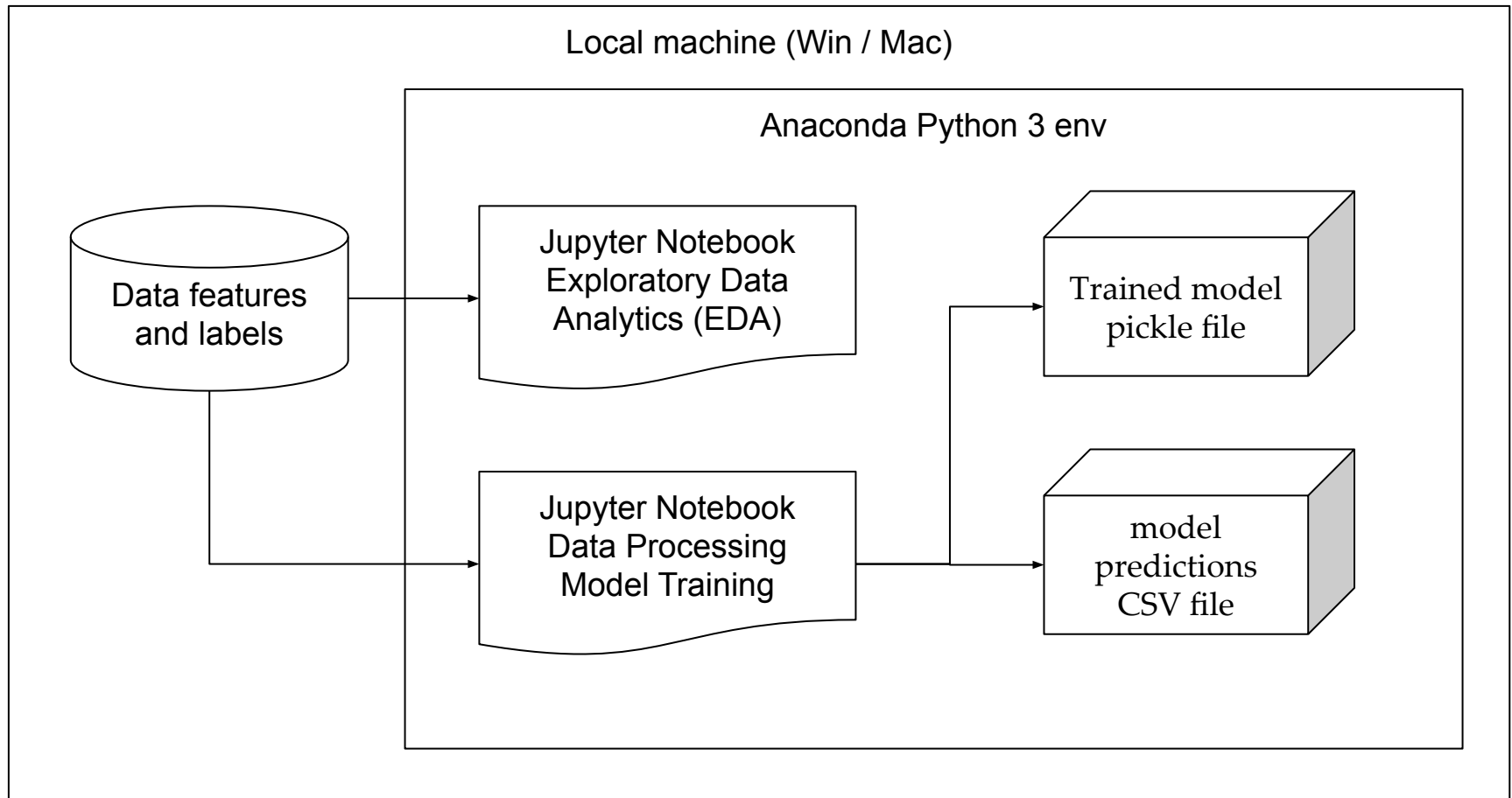
MLOps

Case study

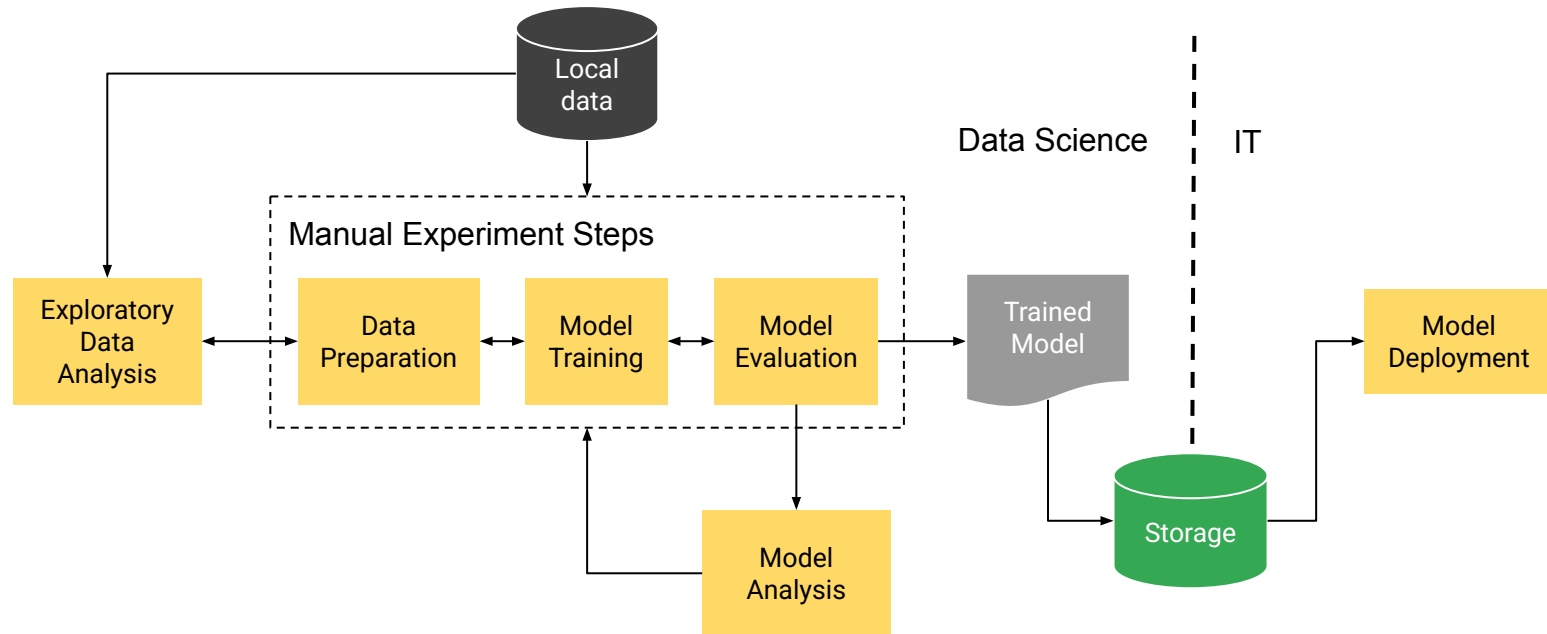
Let's evaluate the earlier student's presentation across the lens of the topics in this module.

Where are you today?

Machine learning project or assignment



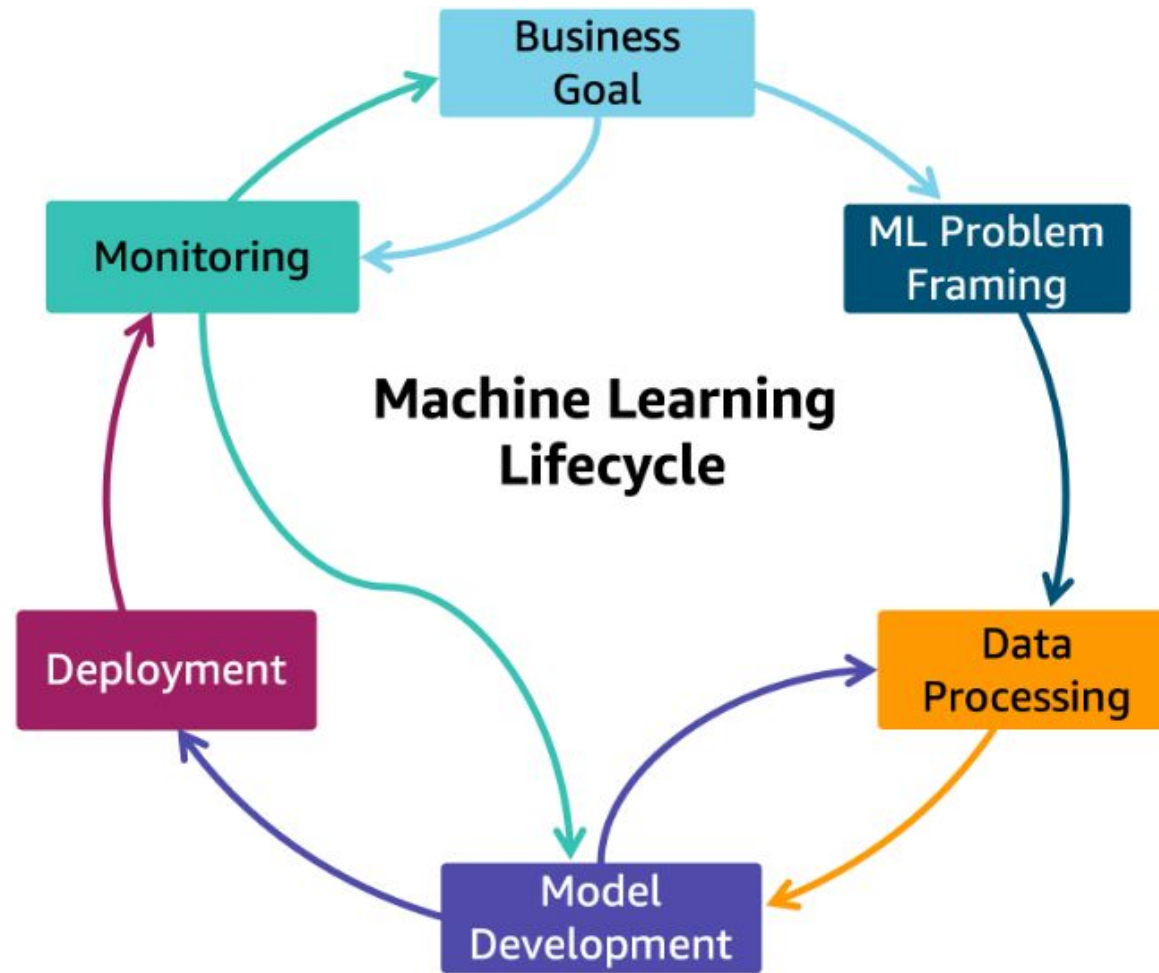
Where are you today?



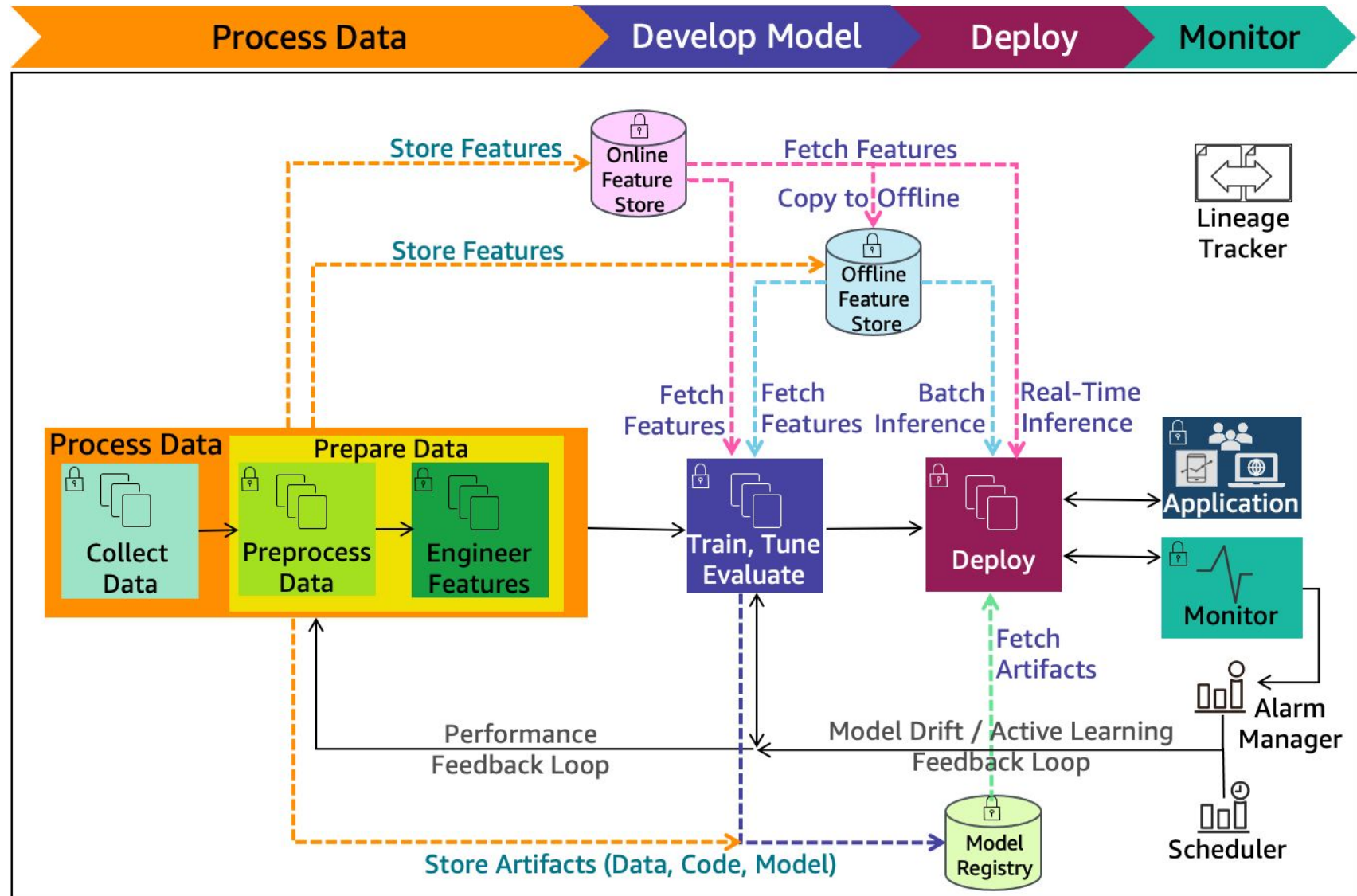
Where are you today?



Machine Learning Lifecycle

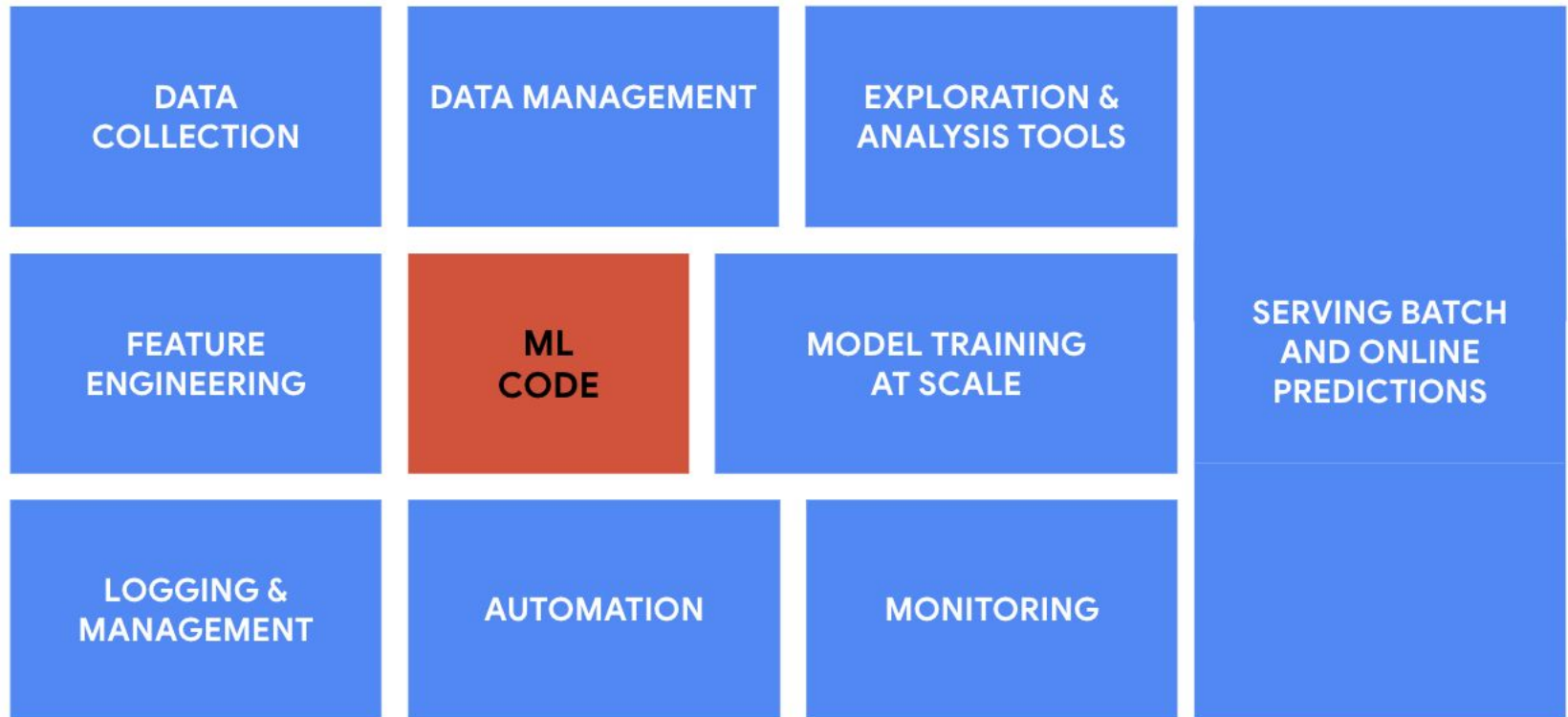


Machine Learning Lifecycle



Challenges of Operationalizing ML

Real-world ML System



ML Deployment is Difficult

“More than 87% of data science projects never make it into production”

- Multiple studies and surveys -

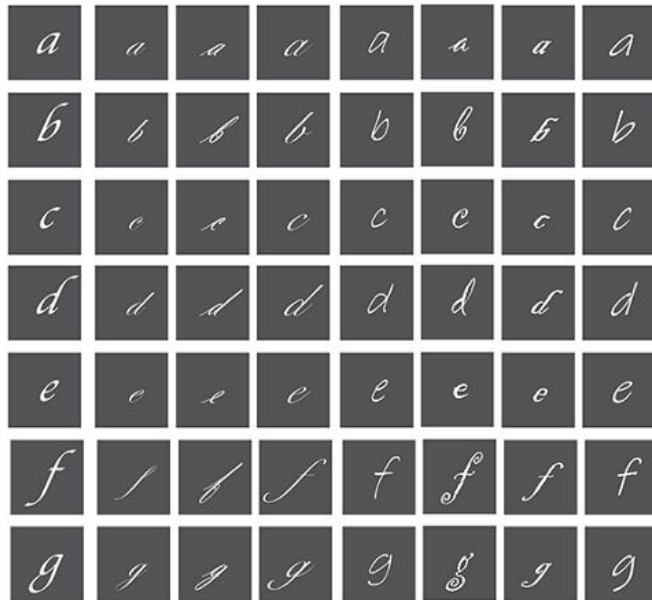
Challenges

All the software engineering concerns

- Real-time or Batch
- Compute resources (CPU/GPU/Memory)
- Latency, throughput (QPS)
- Logging
- Monitoring
- Compliance
- Security and privacy

Challenges

Avoiding training-serving skews that are due to **inconsistencies in data** and in runtime dependencies **between training environments and serving environments.**



Training data



Production data

Source: <https://towardsdatascience.com/machine-learning-in-production-why-you-should-care-about-data-and-concept-drift-d96d0bc907fb>

Challenges

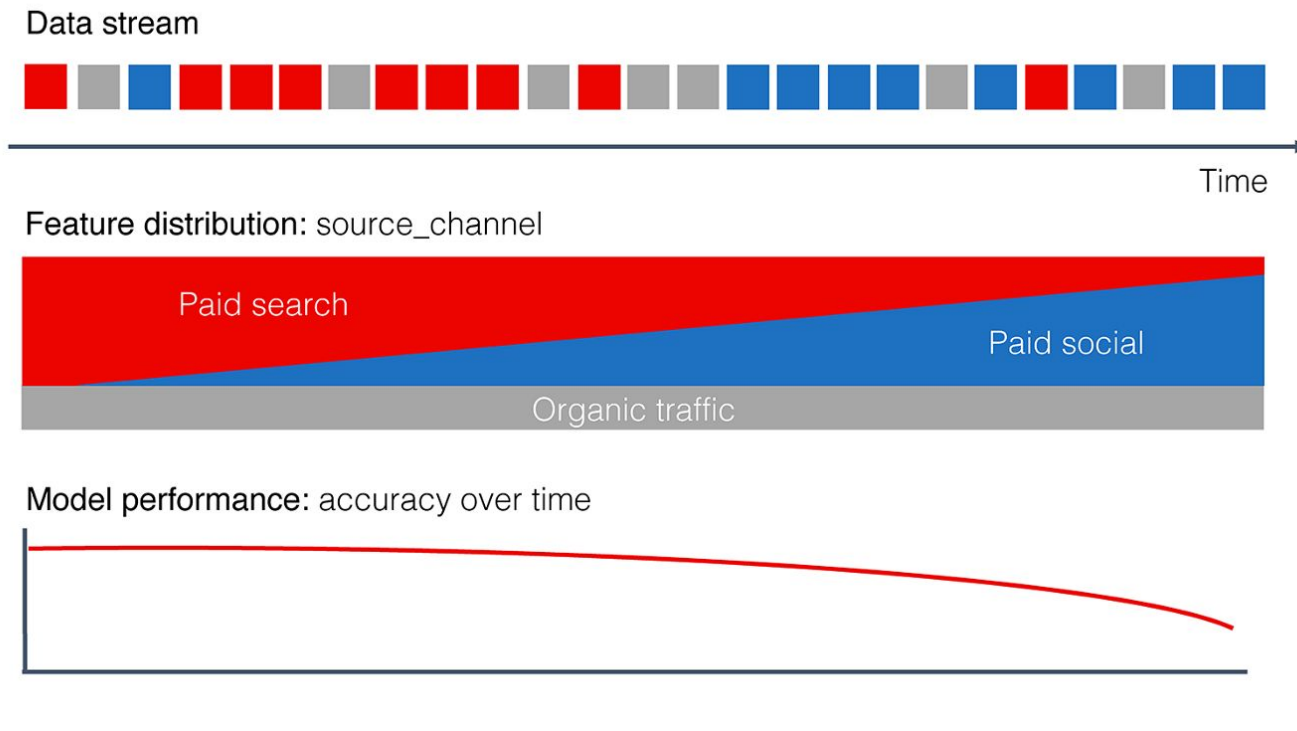
Performing ongoing experimentation of new data sources, ML algorithms, hyperparameters, and then **tracking** these **experiments**.

☐	ID	Owner	Created	Finished	Runnin...	State	{string} Tags	number valid_auc last
☐	CRED-145	 jakub-czakon	08 SEP 2019	08 SEP 2019	2min	SUCCEEDED	lgbm × report ×	0.775489
☐	CRED-110	 jakub-czakon	08 JUL 2019	08 JUL 2019	< 1 min	SUCCEEDED	lgbm ×	0.775489
☐	CRED-108	 jakub-czakon	27 JUN 2019	27 JUN 2019	18min	SUCCEEDED	lgbm ×	0.772611
☐	CRED-93	 kamil	10 JUN 2019	10 JUN 2019	1min	SUCCEEDED	lgbm × features_v1 ×	0.773536
☐	CRED-92	 kamil	10 JUN 2019	10 JUN 2019	< 1 min	SUCCEEDED	lgbm × features_v1 ×	0.765647
☐	CRED-91	 kamil	10 JUN 2019	10 JUN 2019	< 1 min	SUCCEEDED	lgbm × features_v1 ×	0.767254
☐	CRED-89	 kamil	10 JUN 2019	10 JUN 2019	< 1 min	SUCCEEDED	lgbm × features_v1 ×	0.772704
☐	CRED-85	 jakub-czakon	10 JUN 2019	10 JUN 2019	< 1 min	SUCCEEDED	lgbm × features_v1 × production ×	0.775187
☐	CRED-81	 jakub-czakon	07 JUN 2019	07 JUN 2019	1min	SUCCEEDED	lgbm × features_v0 × production ×	0.774266

Source: <https://neptune.ai/blog/ml-experiment-tracking>

Challenges

Maintaining the **performance of models** by continuously retraining them on fresh data.

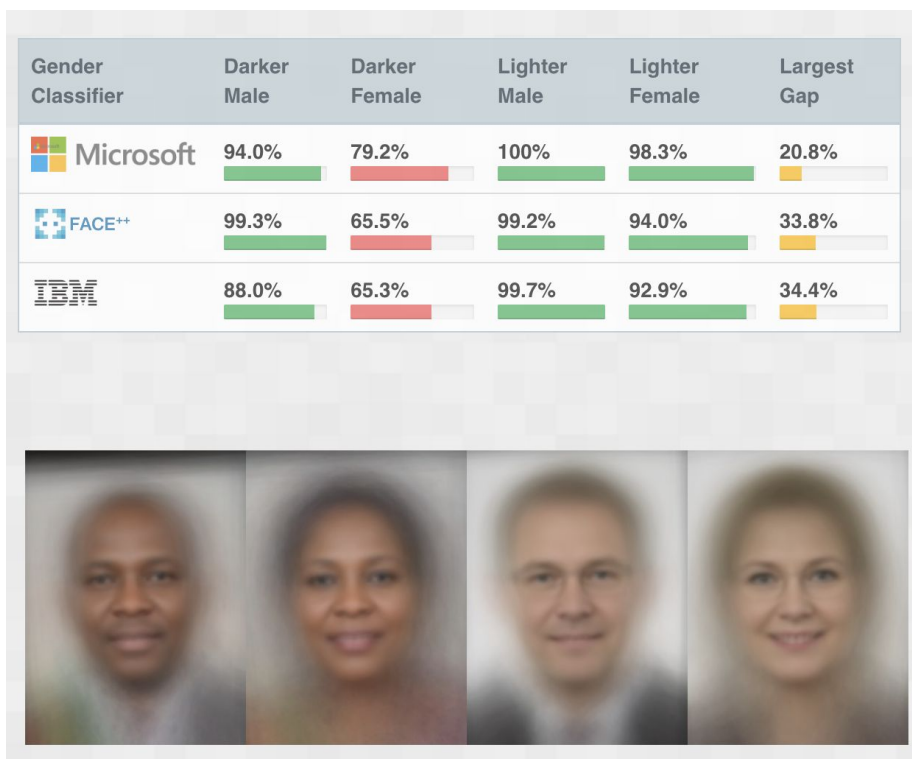


Source: <https://towardsdatascience.com/machine-learning-in-production-why-you-should-care-about-data-and-concept-drift-d96d0bc907fb>

Challenges

Handling concerns about **model fairness, ethics** and **adversarial attacks**.

Since ML models use previously seen data to make decisions, they can worsen issues that already exist in data



Source: <http://gendershades.org/overview.html>

Requirements for Production ML Systems

Reliability

- The system should continue to perform the correct function at the desired level of performance even in the face of adversity (hardware or software faults, and even human error).
- “Correctness” might be difficult to determine for ML systems.
- With traditional software systems, you often get a warning, such as a system crash or runtime error or 404. However, ML systems may fail silently.

Scalability

- There are multiple ways that an ML system can grow
 - Model complexity: a single LR v.s. 100M-parameter neural network
 - Traffic volume: 1000 prediction/day v.s. 1M prediction/day
 - Model counts: 1 model v.s. 1000 models
- Need reasonable ways of dealing with that growth
- An indispensable feature in many cloud services is **autoscaling**
 - automatically scaling up and down the number of machines depending on usage
- However, handling growth isn't just resource scaling, but also **artifact/metadata management**

Maintainability

- ML-enabled system is a complex undertaking that combines data engineering, ML engineering, and application engineering tasks
- Many people who will work on an ML system: ML engineers, DevOps engineers, Data analyst, Data scientist
 - Docker file
 - Model source code (.py, .java)
 - Data processing source code (.sql)
 - Notebooks (.ipynb)
 - Configs (.yaml, .ini)
 - Pipeline specs (.json)
 - Scripts (.sh)
- Different contributors should be able to work together

Adaptability

- Machine learning problem is dynamic
 - business requirements
 - external environment
 - ...
- Data distributions may shift accordingly
- Because ML systems are part code, part data, and data can change quickly, ML systems need to be able to evolve quickly.

In an IT system

Behavior of system is defined by **Code**

In an AI/ML System

Behavior of system is defined by **Data**

Iterative Process

- Developing an ML system is an iterative and, in most cases, never ending process
- The speed of going through an iteration of ML lifecycle determines the speed of realizing the value of ML
- ML system should support developer with process and tools for fast experiment and iteration
- Continuous training, continuous integration and continuous deployment

Machine Learning Pipeline



Data Processing

Data Processing

Data Collection

Data Preparation (Wrangling during Interactive Data Analysis) Leverage **Visualization** for Exploratory Data Analysis (EDA)

Data Collection

Label

Ingest (Streaming, Batch)

Aggregate

Data Preprocessing

Clean (Replace, Impute,
 Remove Outliers, Duplicates)

Partition (Train, Validate, Test)

Scale(Normalize, Standardize)

Unbias, Balance
 (Detection & Mitigation)

Augment

Feature Engineering

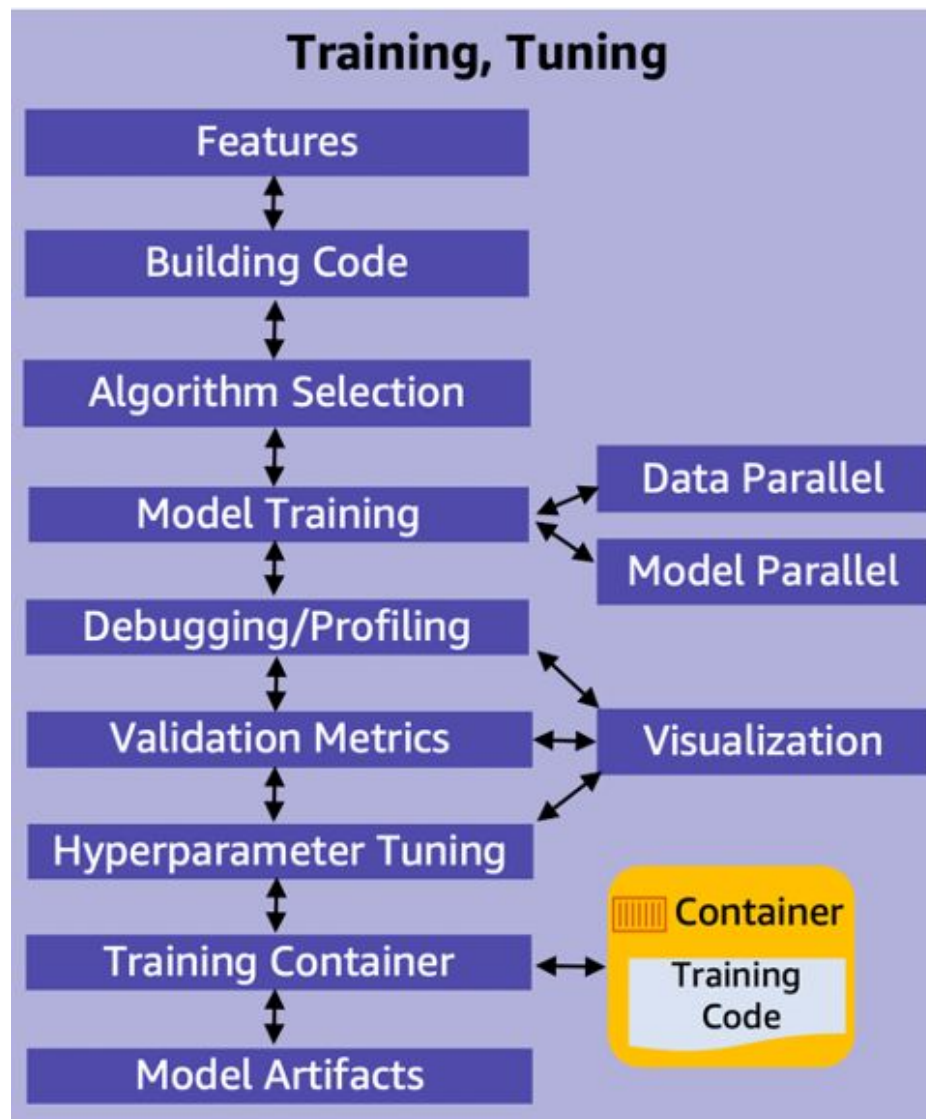
Feature Selection

Feature Transformation

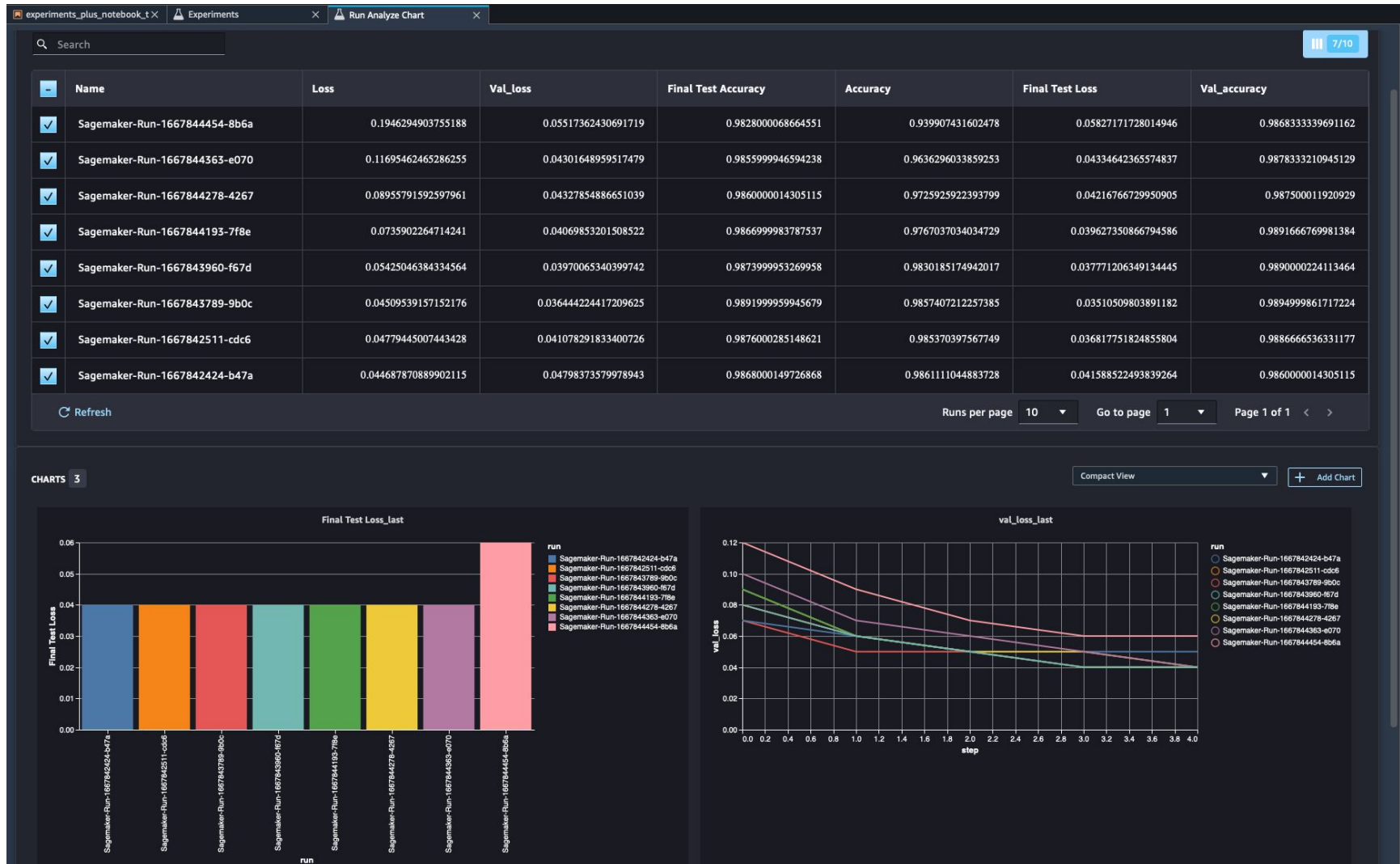
Feature Creation
 (Encoding, Binning)

Feature Extraction
 (Automated in Deep Learning)

Model Training



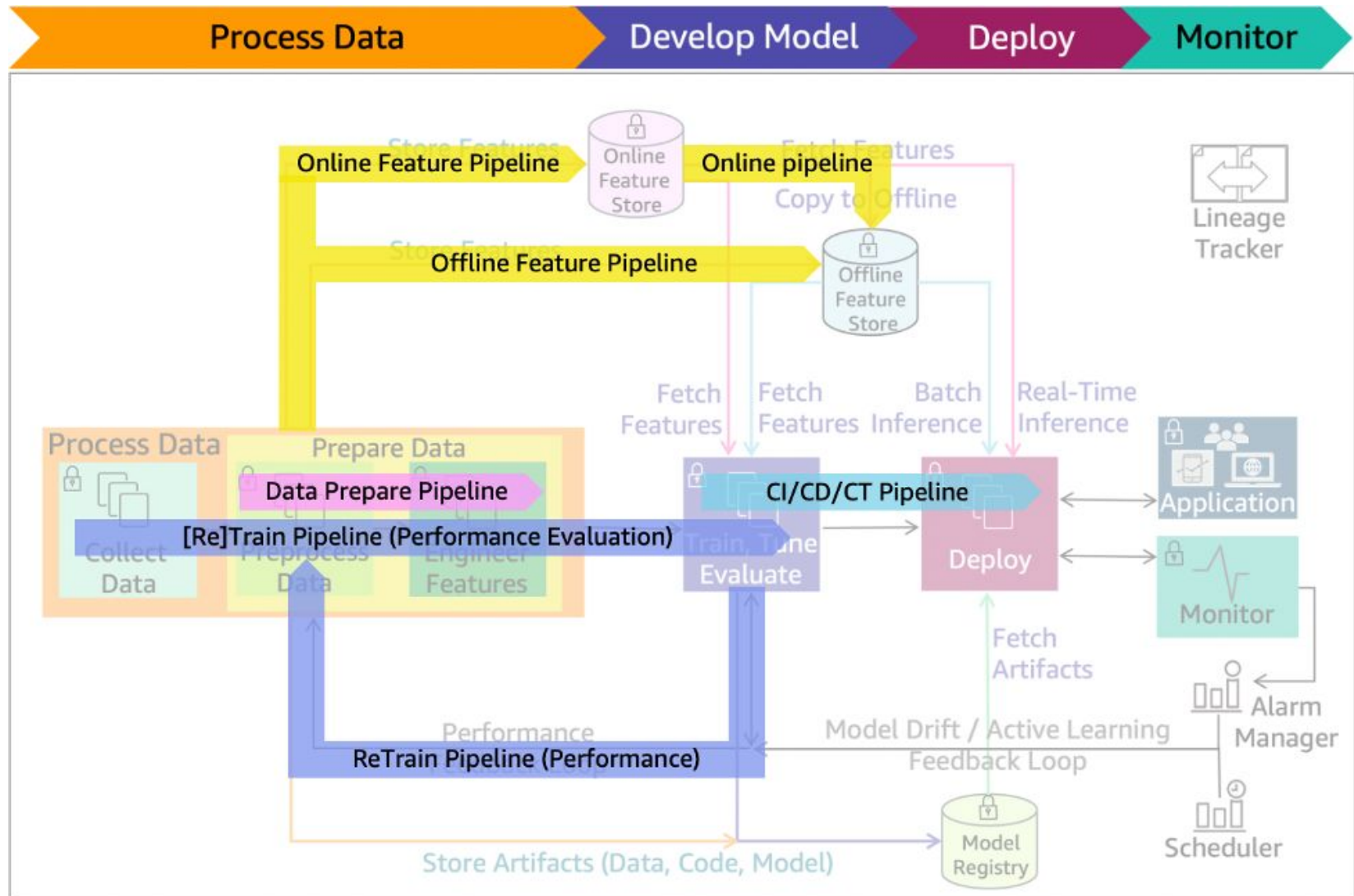
Model Evaluation



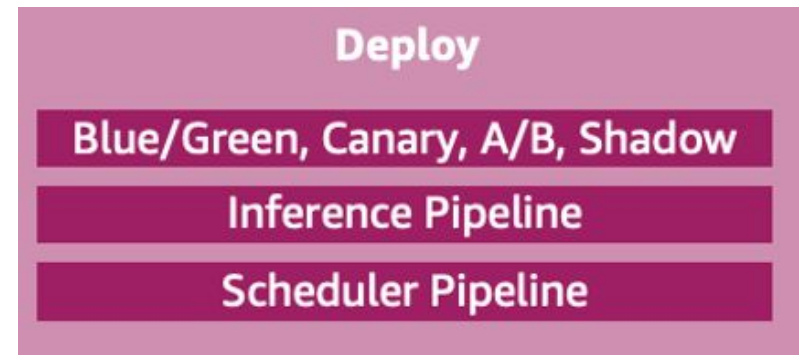
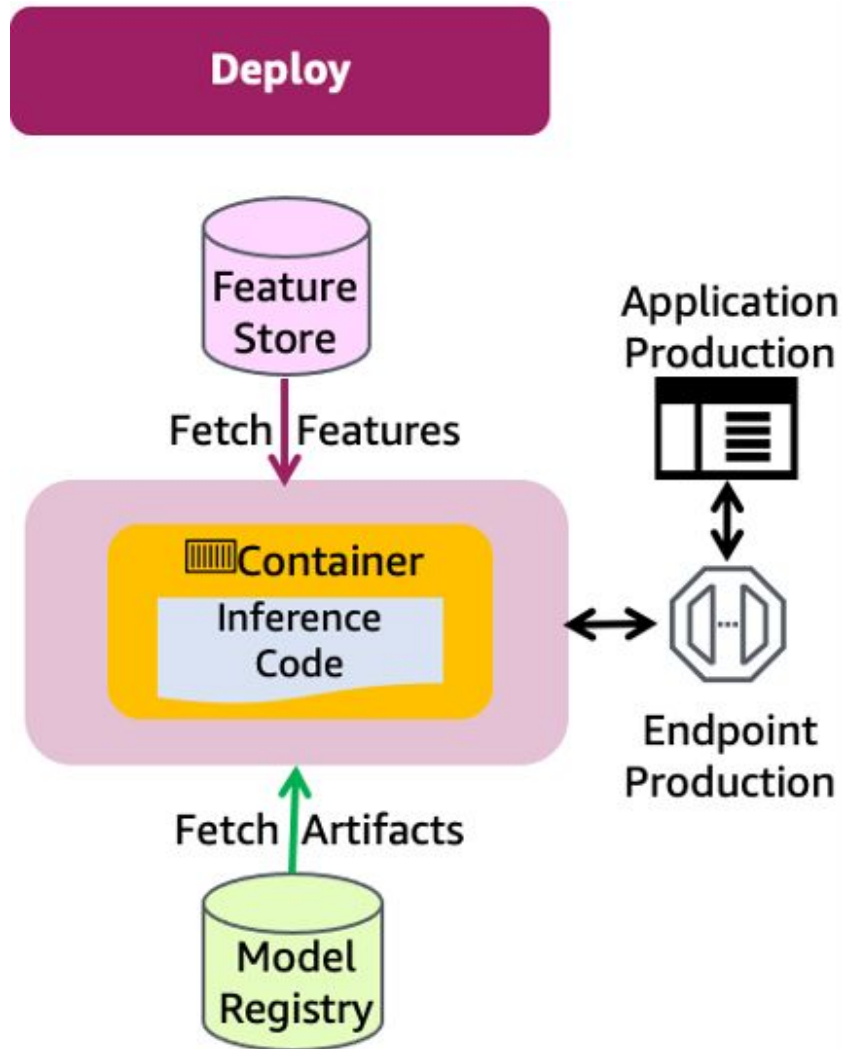
Model Evaluation and Validation

- Select the appropriate performance metric to evaluate models based on the business problem it is solving
- Evaluate the model's performance on these metrics across datasets and time
- Validate a model by setting up value thresholds and change thresholds based on the desired metrics
 - Accuracy, precision, recall
 - QPS
- Change threshold is useful to gate the candidate model by checking whether the corresponding metric is larger/smaller than that of a baseline model; Absolute change vs Relative change.

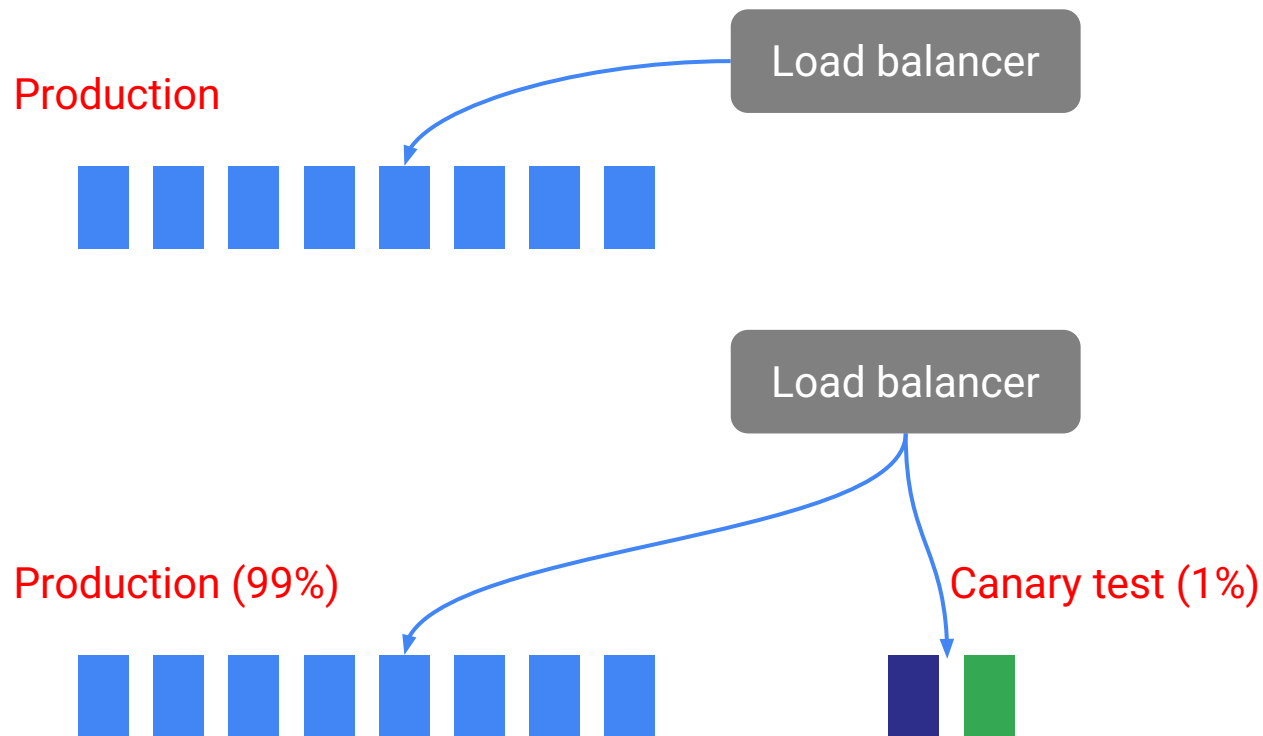
Model Evaluation



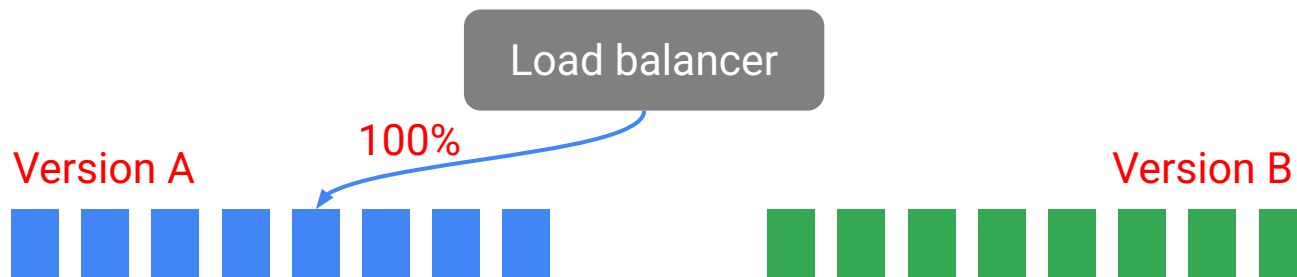
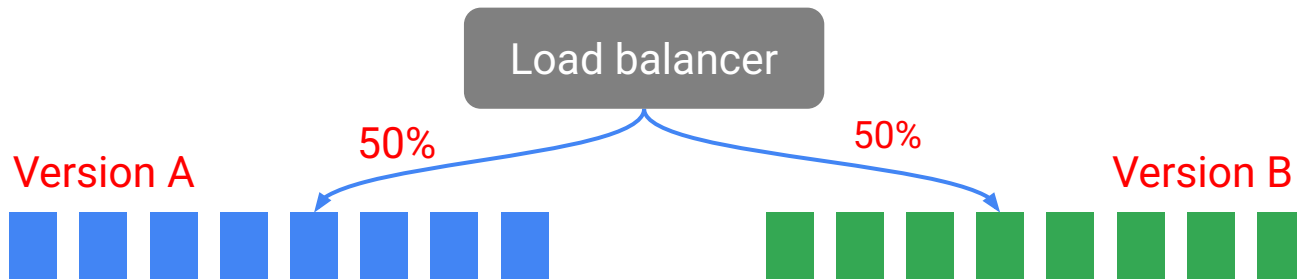
Model Deployment



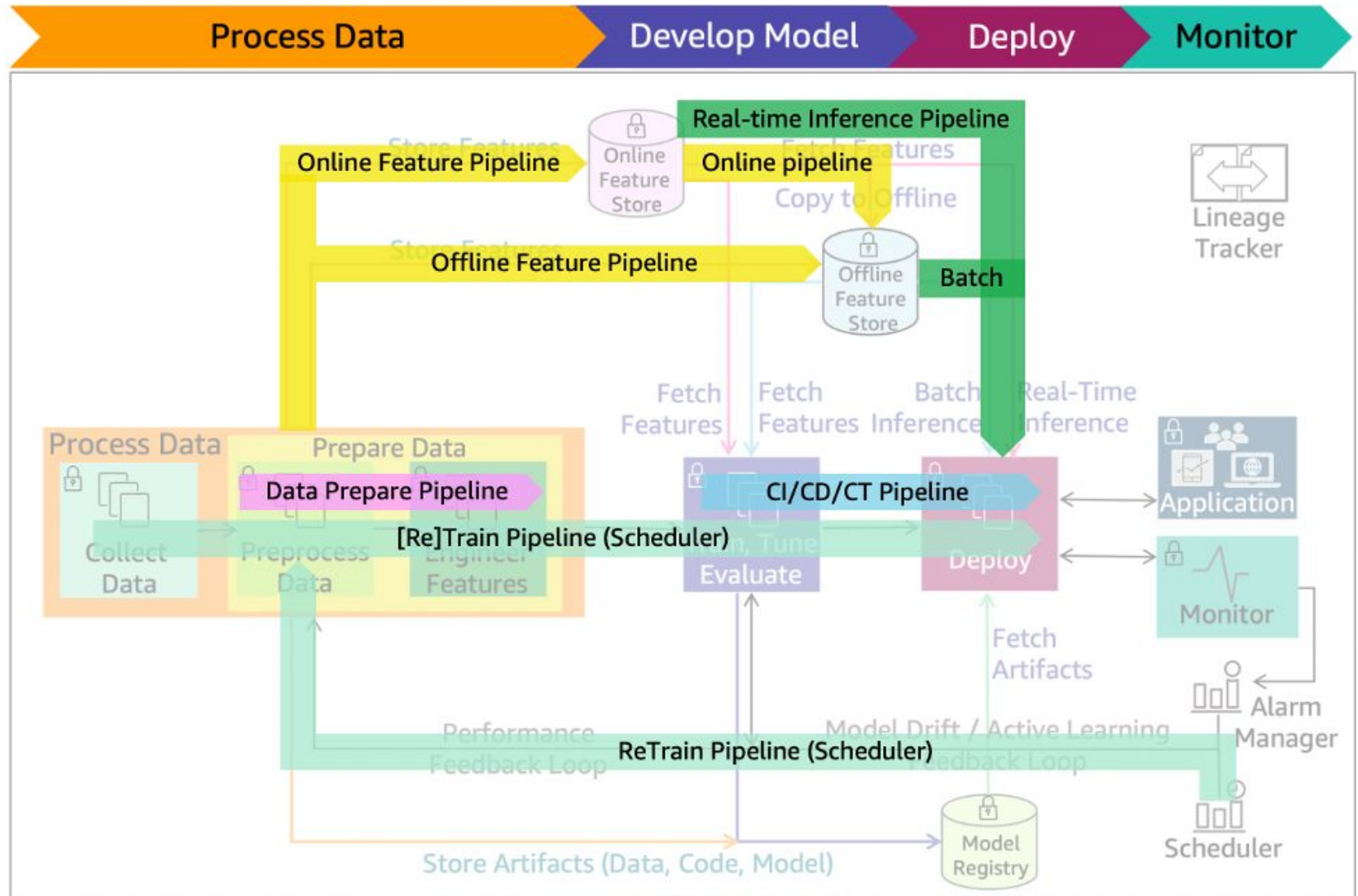
Model Deployment



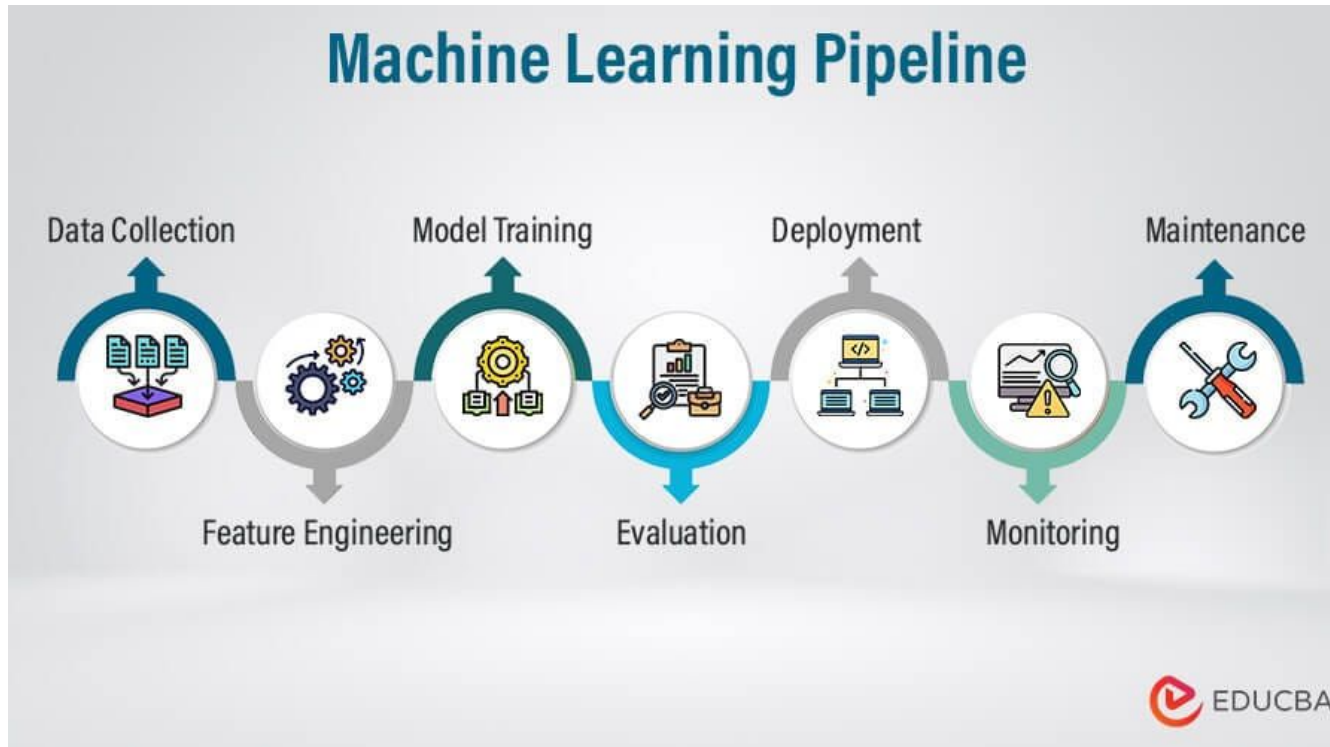
Model Deployment



Model Deployment



Pipeline



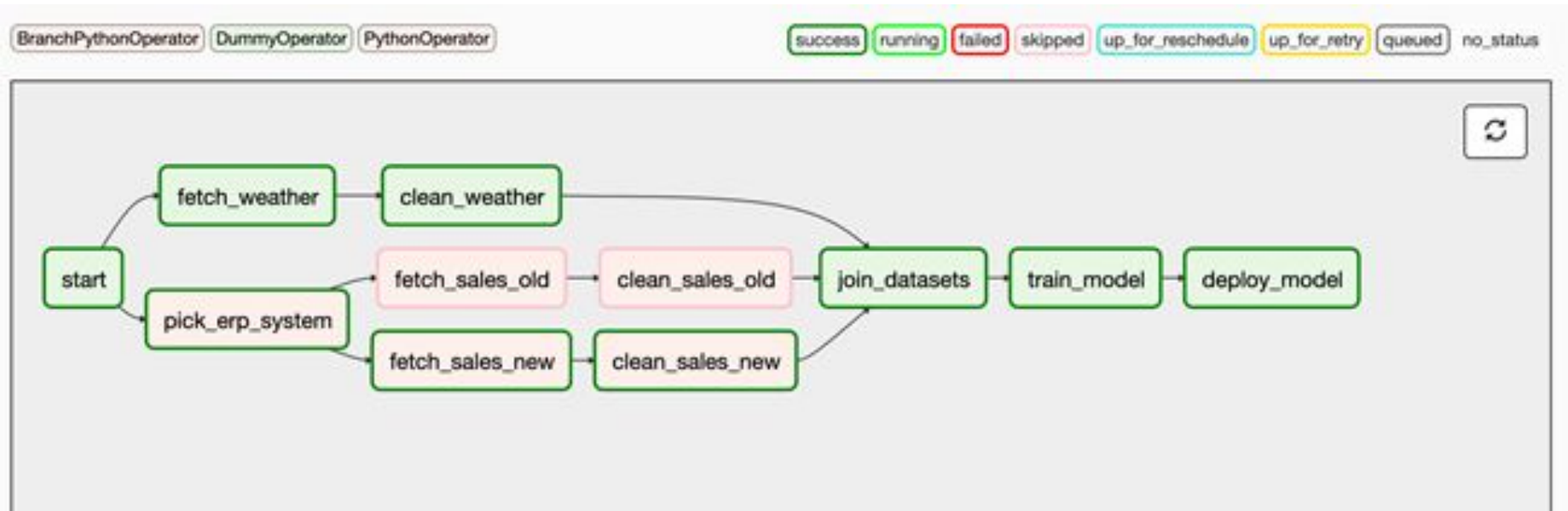
Automation of the model life cycle steps
Prevention of bugs
Useful paper trail
Standardization
Free up development time for data scientists

Source:
<https://medium.com/@surajagrahari330/machine-learning-pipeline-from-raw-data-to-insightful-models-80c54914b7a9>

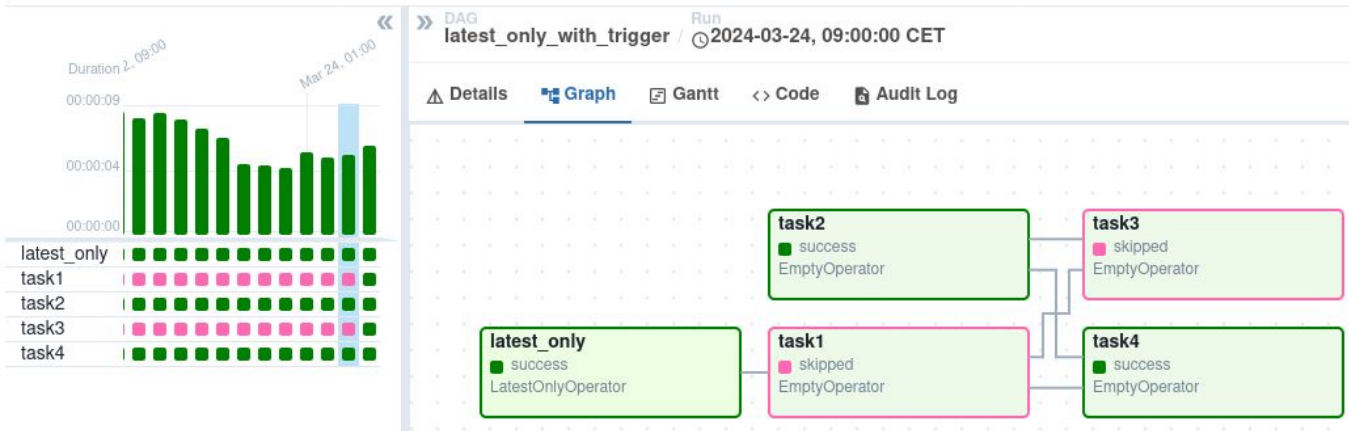
Pipeline Orchestration

- Pipeline tools manage the flow of tasks through a graph representation of the task dependencies.
- The pipeline steps are directed, which guarantees that the path of execution is clearly defined by the tasks' dependencies
- Triggering of the pipeline
 - On-demand
 - On-schedule
 - On-availability of new data
 - On-model performance degradation
 -

Pipeline Orchestration



Press **shift** + **/** for Shortcuts



source:
<https://airflow.apache.org/docs/apache-airflow/stable/core-concepts/dags.html>

Model Monitor

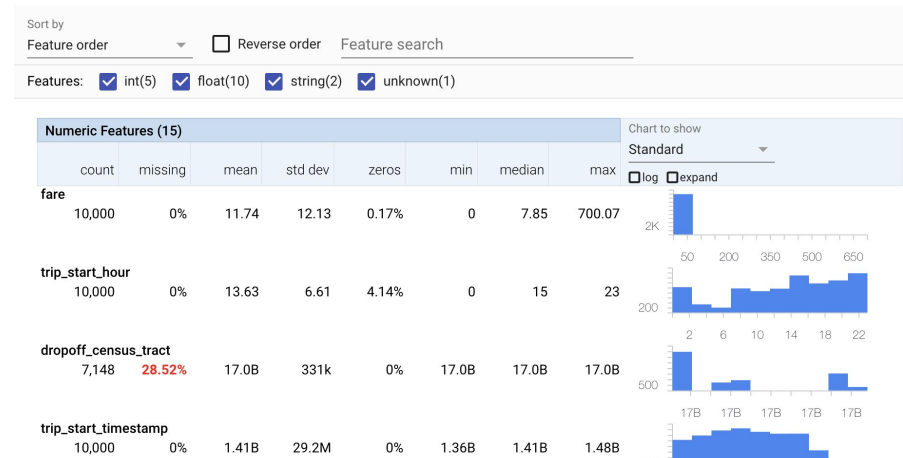
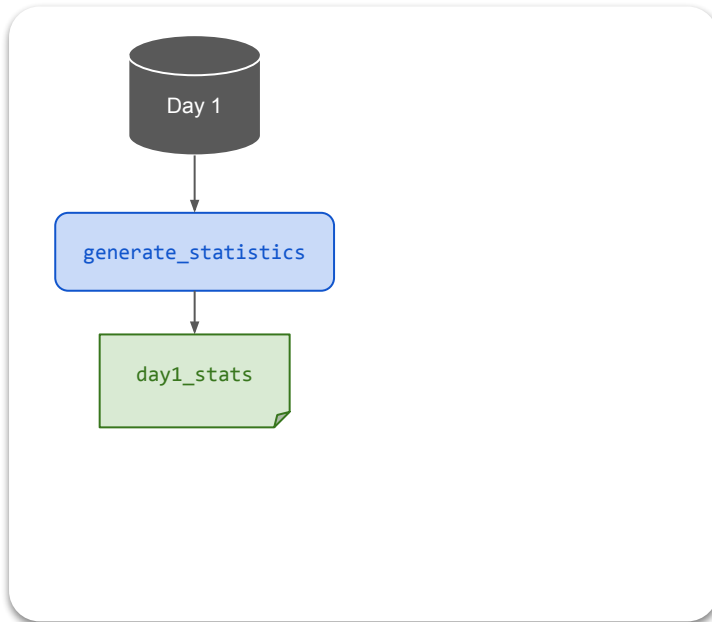
Monitor

Model Explainability

Detect Drift

Model Update Pipeline

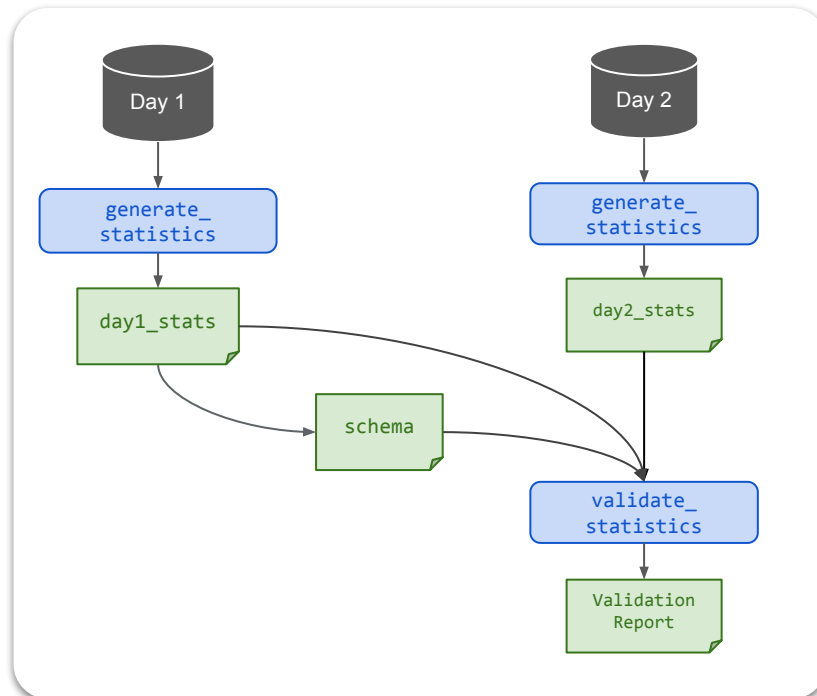
Data Validation



Compute and visualize descriptive statistics

Source: <https://www.tensorflow.org/tfx/guide>

Data Validation

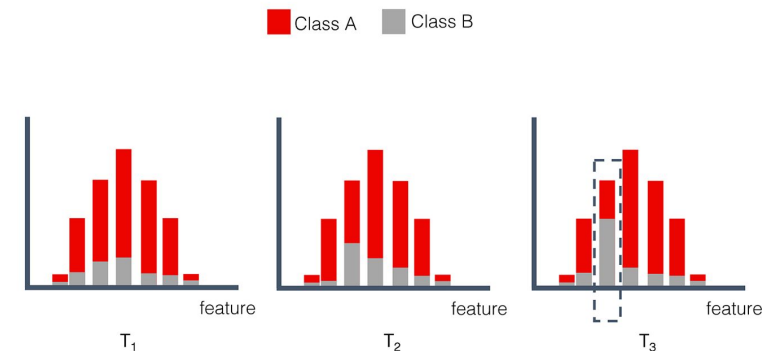
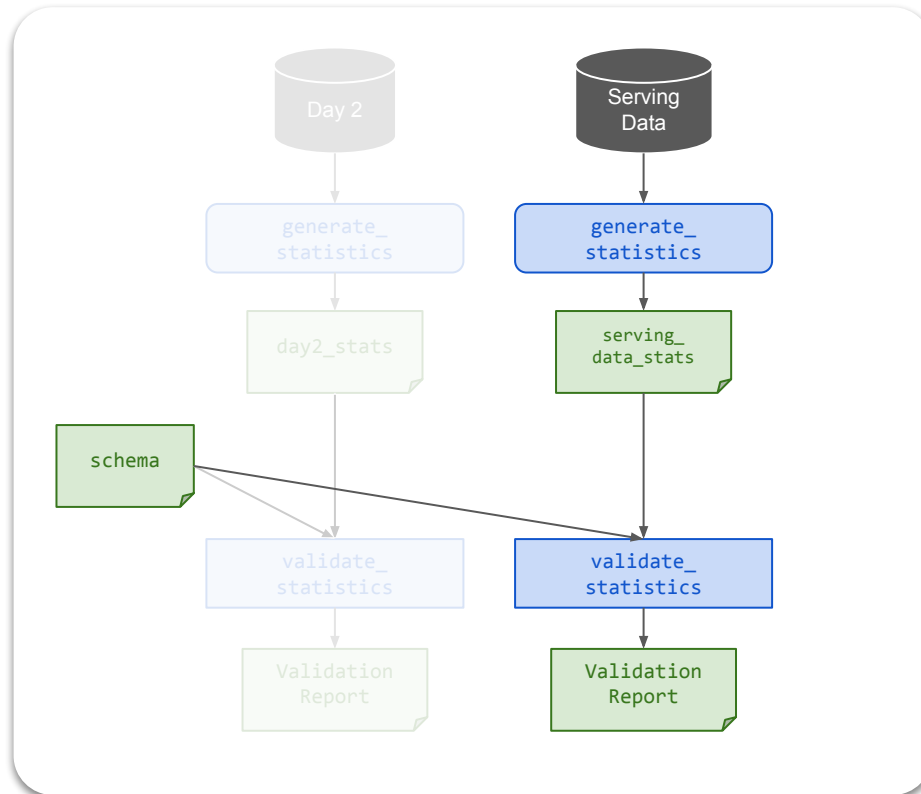


Anomaly short description		Anomaly long description
Feature name		
'payment_type'	Unexpected string values	Examples contain values missing from the schema: Prcard (<1%).
'company'	Unexpected string values	Examples contain values missing from the schema: 2092 - 61288 Sbeih company (<1%), 2192 - 73487 Zeymane Corp (<1%), 2192 - Zeymane Corp (<1%), 2823 - 73307 Seung Lee (<1%), 3094 - 24059 G.L.B. Cab Co (<1%), 3319 - CD Cab Co (<1%), 3385 - Eman Cab (<1%), 3897 - 57856 Ilie Malec (<1%), 4053 - 40193 Adwar H. Nikola (<1%), 4197 - Royal Star (<1%), 585 - 88805 Valley Cab Co (<1%), 5874 - Sergey Cab Corp. (<1%), 6057 - 24657 Richard Addo (<1%), 6574 - Babylon Express Inc. (<1%), 6742 - 83735 Tasha ride inc (<1%).

Schema validation

Source: <https://www.tensorflow.org/tfx/guide>

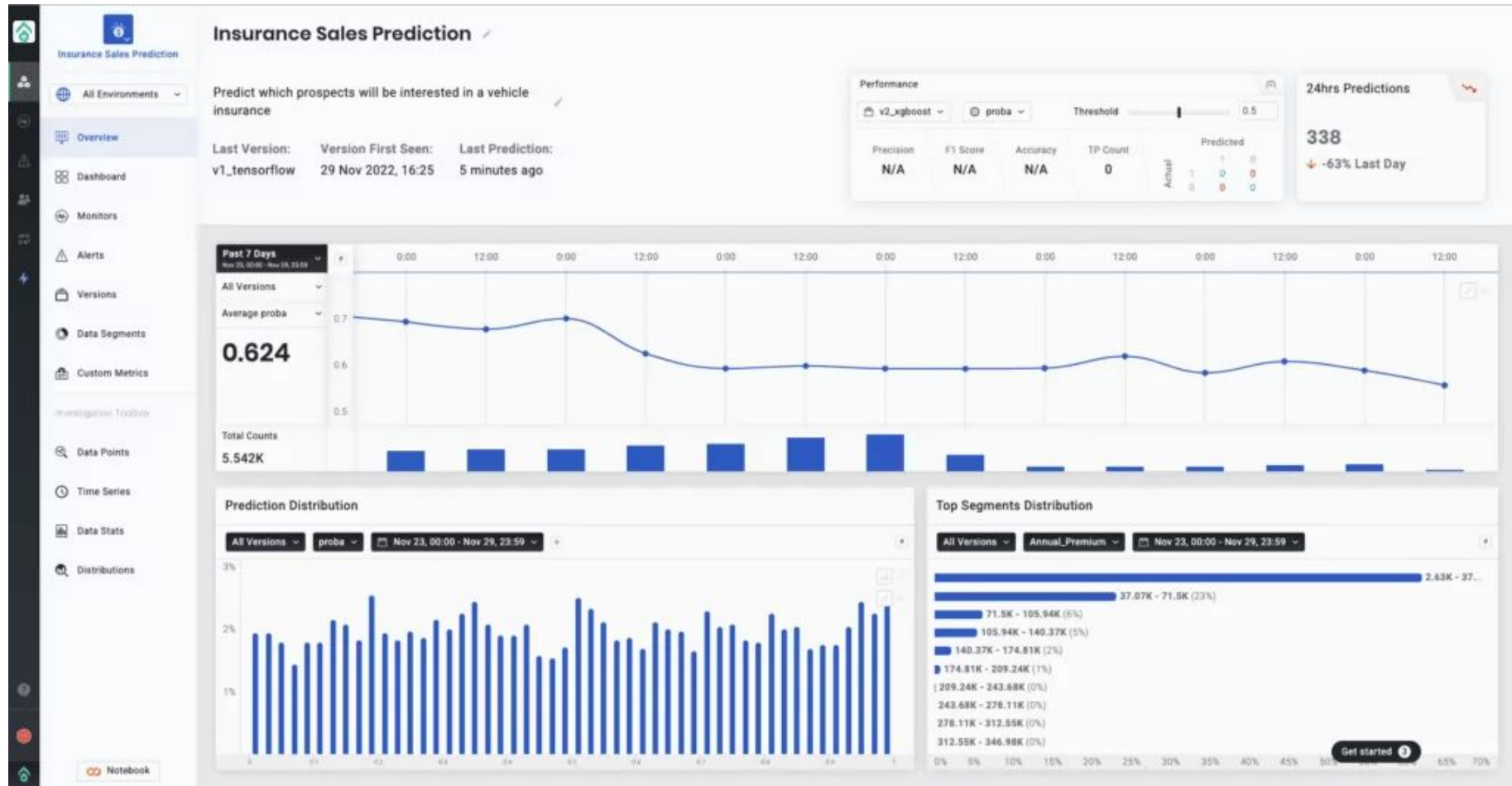
Data Validation



Training/serving skew detection

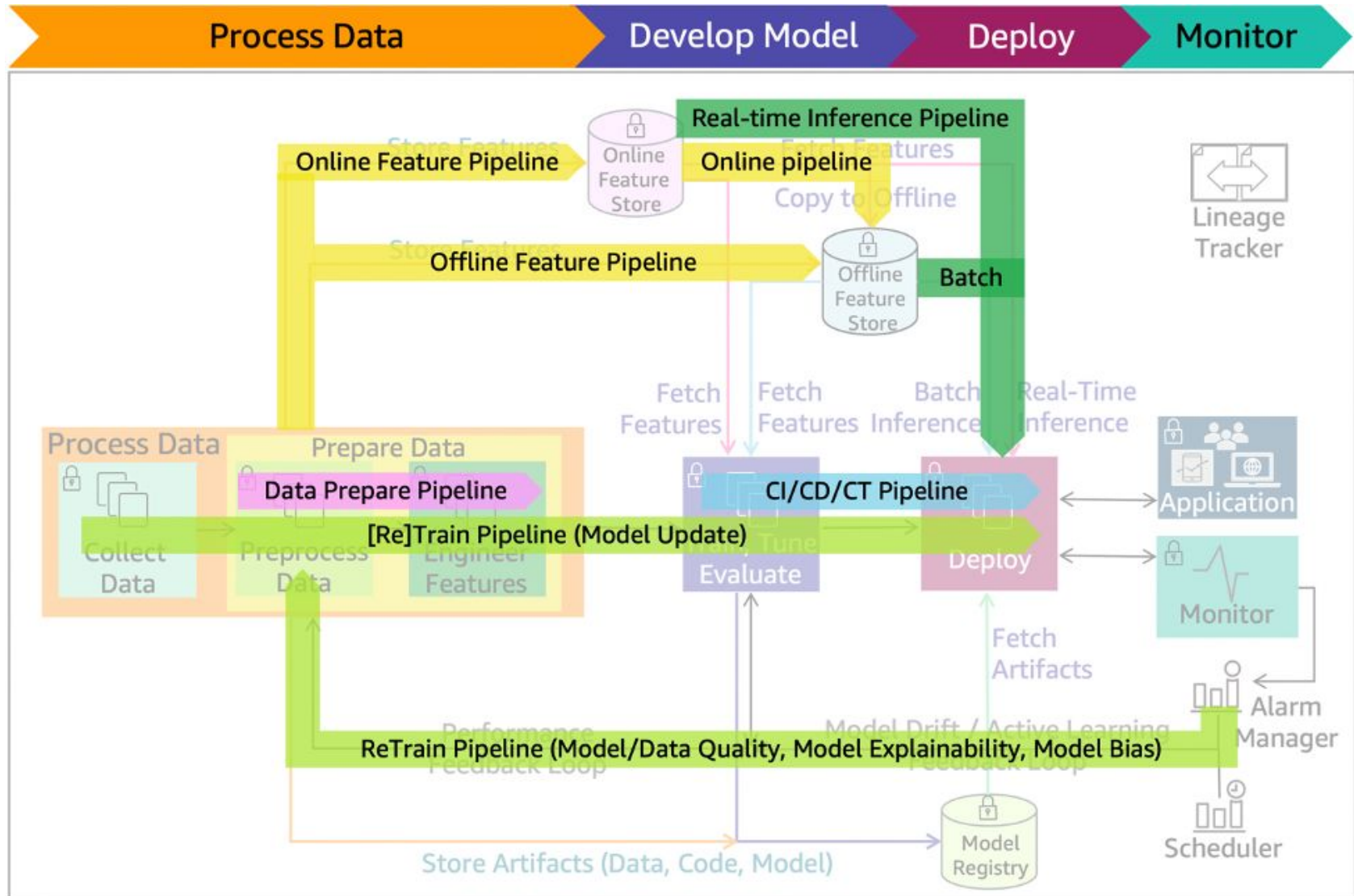
Source: <https://www.tensorflow.org/tfx/guide>

Model Monitoring



Source: <https://www.tensorflow.org/tfx/guide>

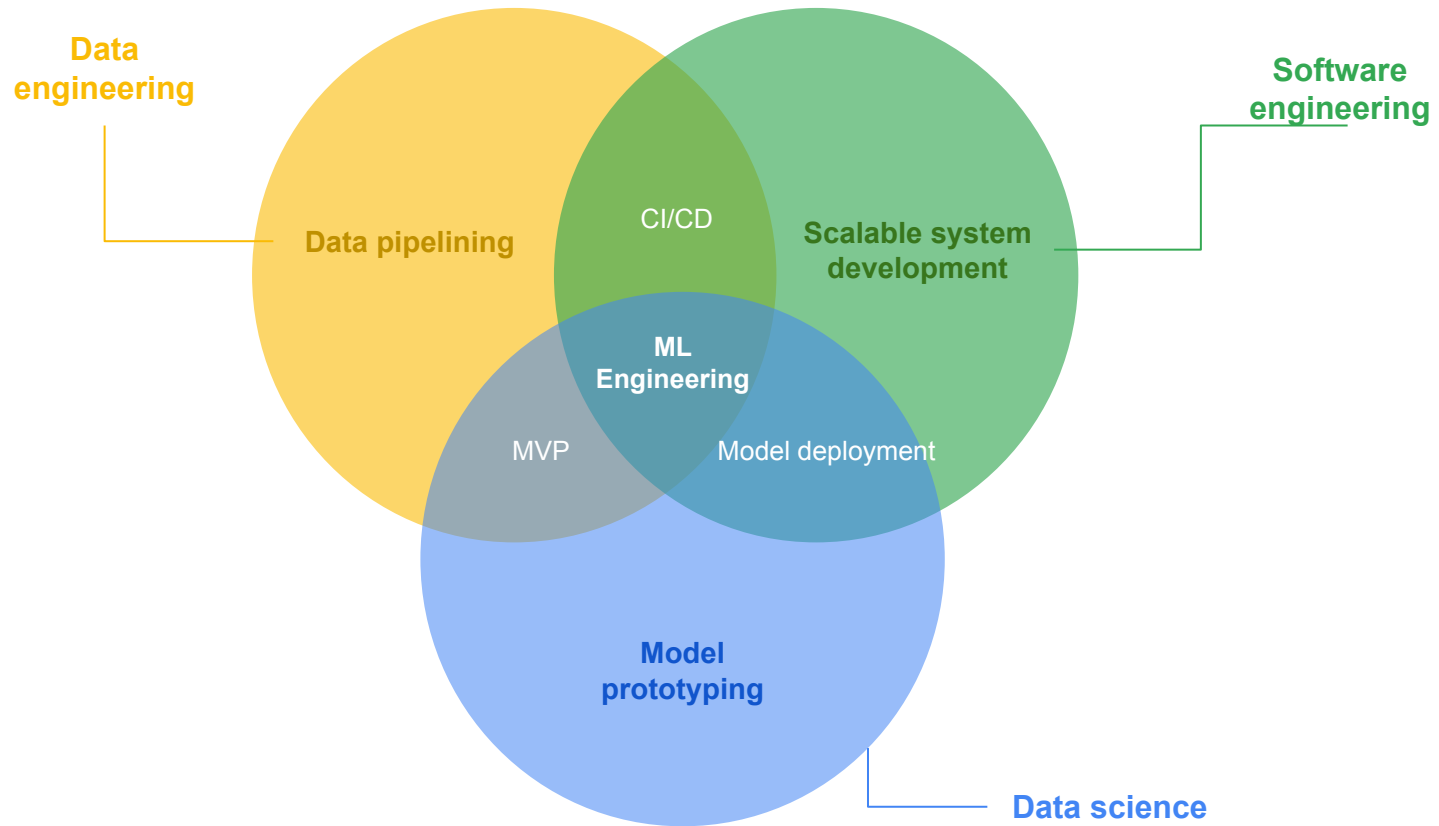
Model Monitoring



MLOps

Practitioners guide to MLOps: A framework for continuous delivery and automation of machine learning

Software Engineering v.s. ML Engineering



What is MLOps

- MLOps is a methodology for ML engineering that unifies ML system development (the ML element) with ML system operations (the Ops element).
- MLOps provides a set of standardized processes and technology capabilities for building, deploying, and operationalizing ML systems rapidly and reliably.
- MLOps supports ML development and deployment in the way that DevOps and DataOps support application engineering and data engineering (analytics).

What is MLOps

- DevOps, DataOps, MLOps
 - When you deploy a web service, you care about resilience, queries per second, load balancing, and so on.
 - When you deploy an ML model, you also need to worry about changes in the data, changes in the model, users trying to game the system, and so on.

- MLOps practices can result in the following benefits
 - Increase productivity, resulting in faster time-to-market for ML projects.
 - Ensure repeatable processes, including how the ML model is trained, evaluated, versioned, and deployed.
 - Increase reliability, performance, scalability, and security of ML systems.
 - Streamlined operational and governance processes.
 - Increased return on investment of ML projects.

MLOps Lifecycle



An aerial night photograph of Singapore. The foreground shows a large, modern building with a brown roof and glass facade, identified by the 'SMU' logo. The background features a dense city skyline with numerous illuminated skyscrapers under a dark sky. The text 'Questions?' is overlaid in a bold, orange font on the left side of the image.

Questions?

Thank You